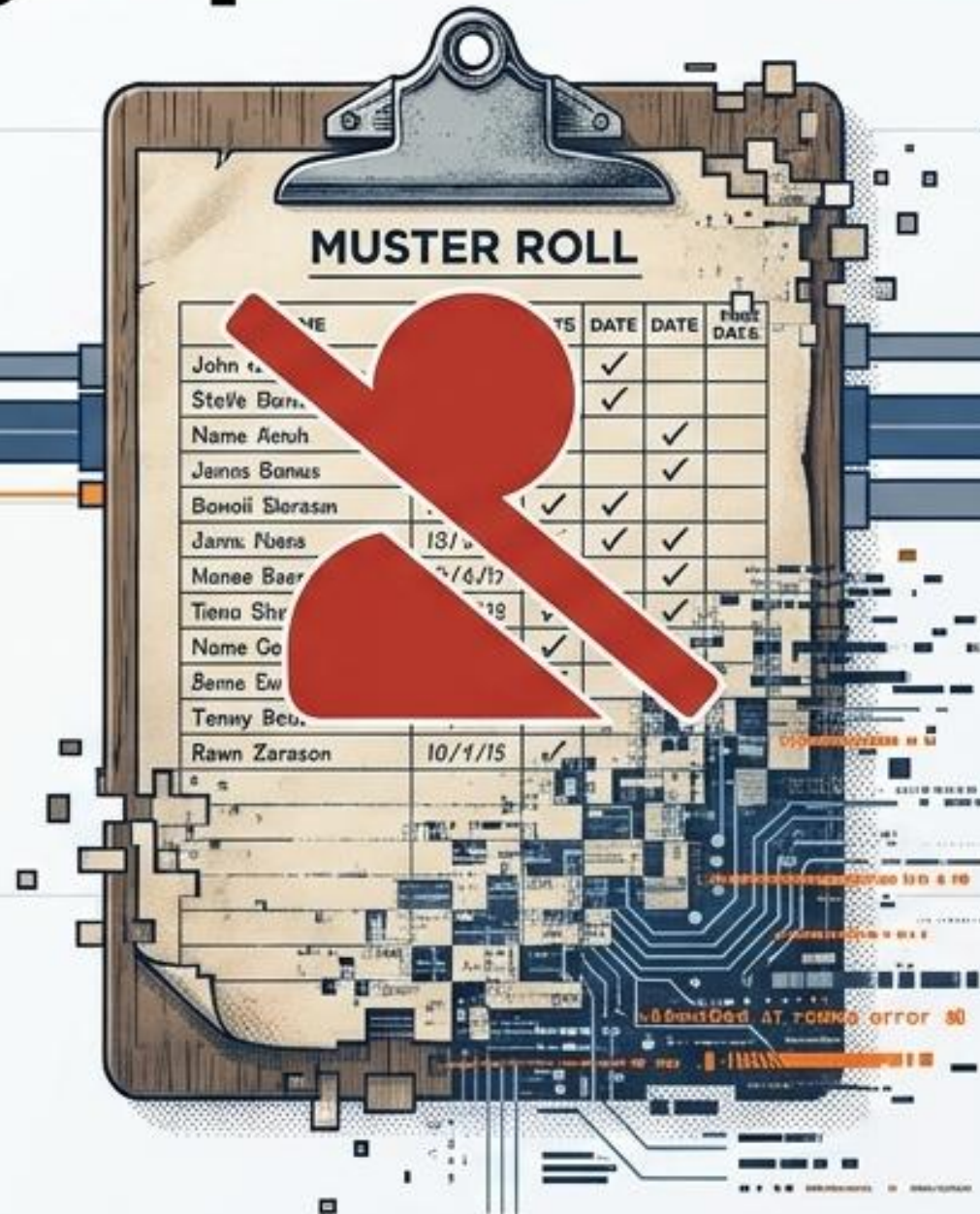


SiteID

Offline-First Worker Identity
& **Wage Protection** for
Indian Construction Sites



Construction wages disappear through a single point of failure



The Core Thesis

DOT, 2013, 1311.1933, 313

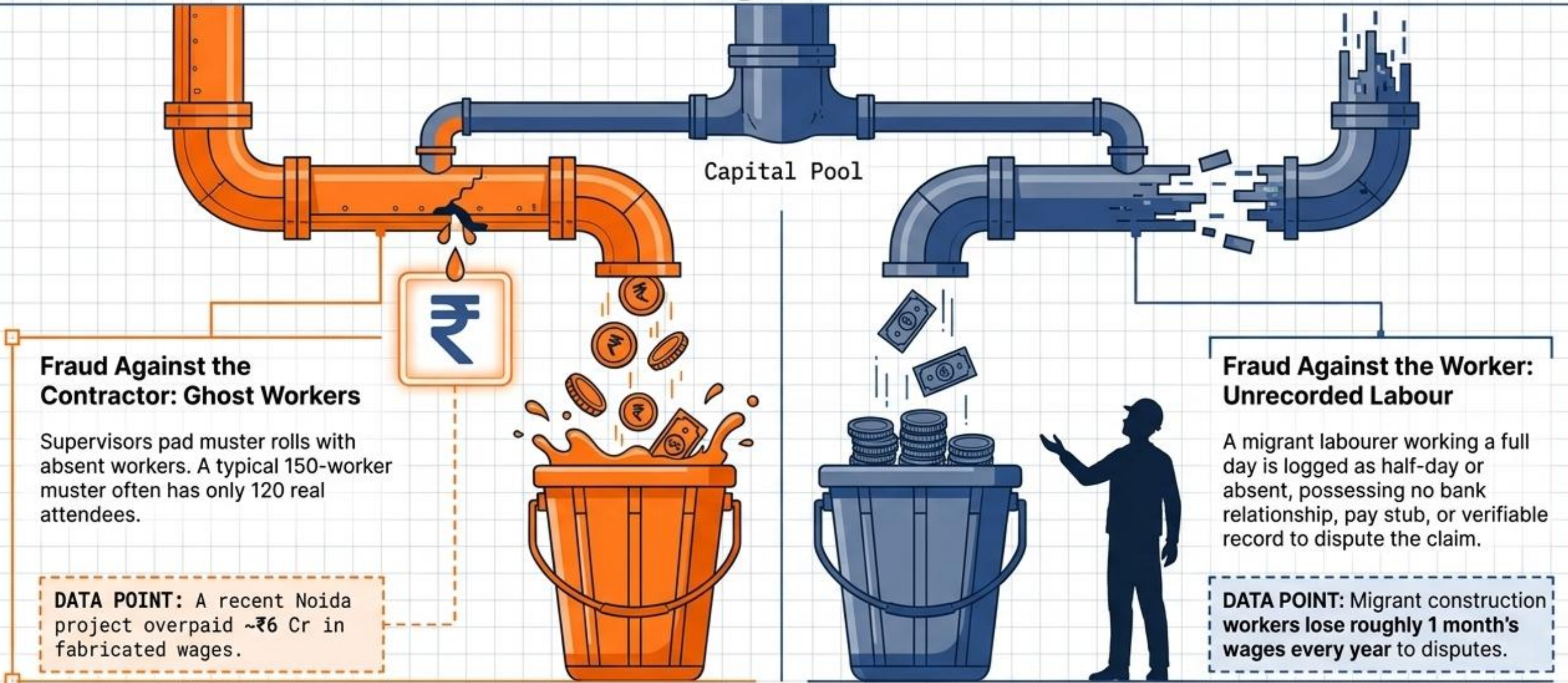
Attendance is currently recorded on paper by individuals who have a direct financial interest in the outcome.

The Flaw

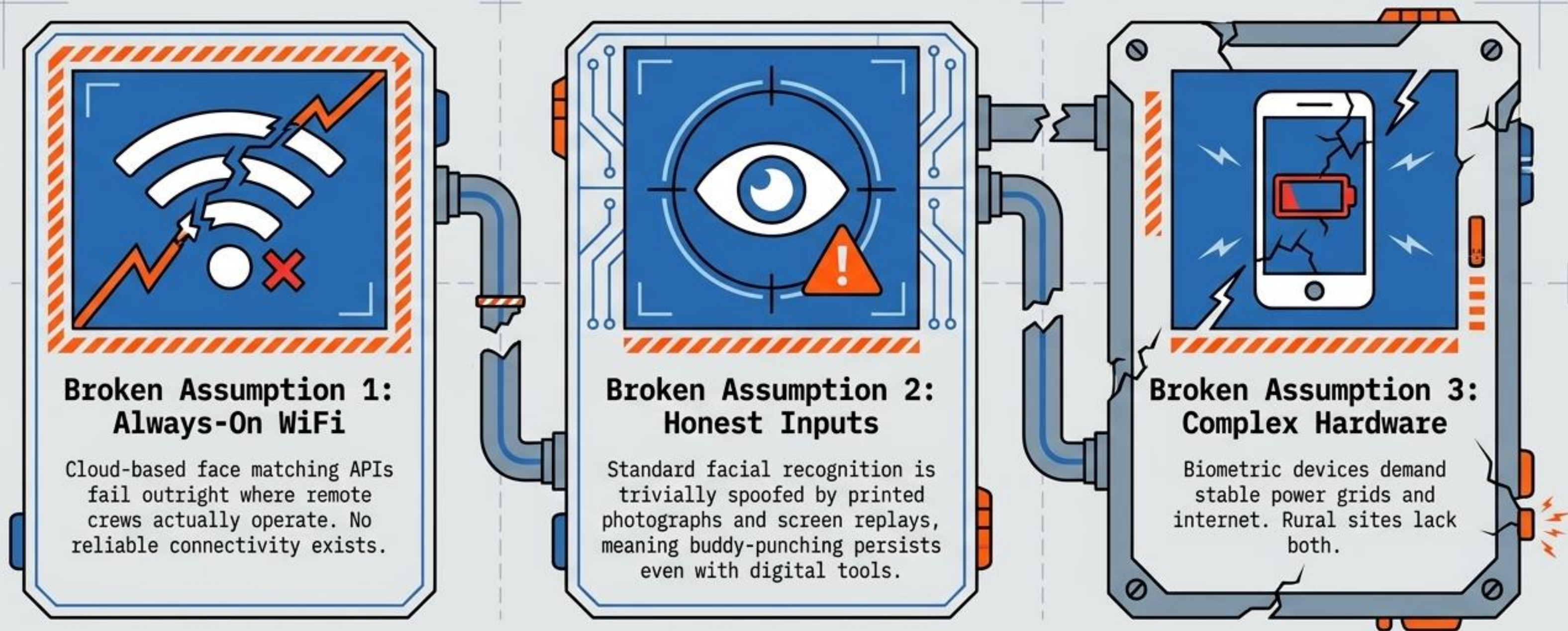
DOT, 2013, 1311.1833, 313

When verification depends entirely on the integrity of the person filing the record, the resulting fraud cuts both ways.

The paper paradigm bleeds capital from the contractor and steals wages from the worker



Rural environments immediately break the assumptions of cloud-based biometrics



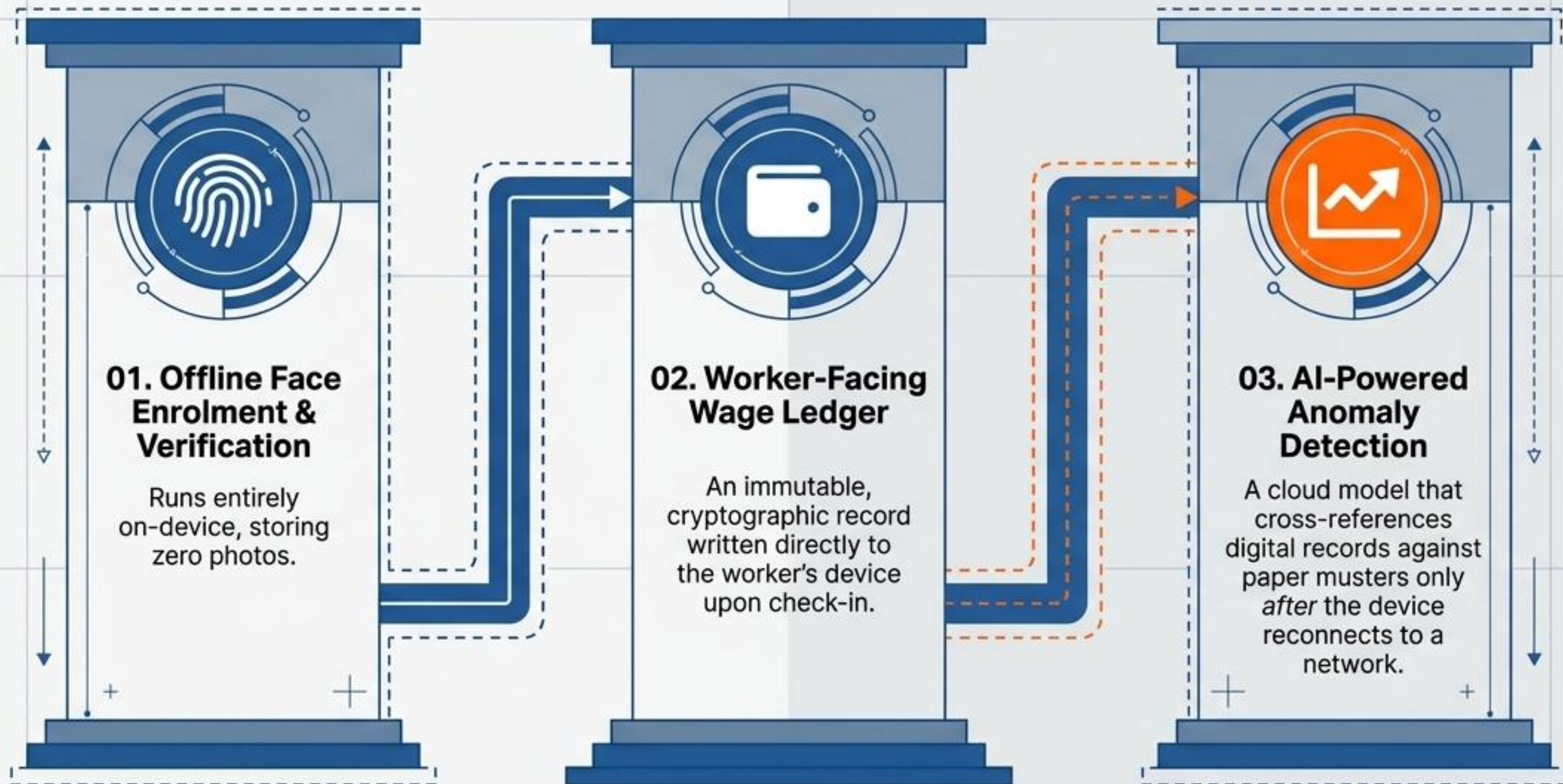
Existing tools solve attendance logging, but ignore offline spoofing and worker empowerment



	Paper Musters	Cloud Biometrics (Truein/SiteSetu)	SiteID (Ours)
Logs Attendance	✓	✓	✓
Works Fully Offline	✓	✗	✓
Defeats Photo Spoofing	✗	✗	✓
Provides Worker-Owned Records	✗	✗	✓

The Market Gap: Competitors build for the owner. We build cryptographic infrastructure for places where physical infrastructure fails, guaranteeing trust without connectivity.

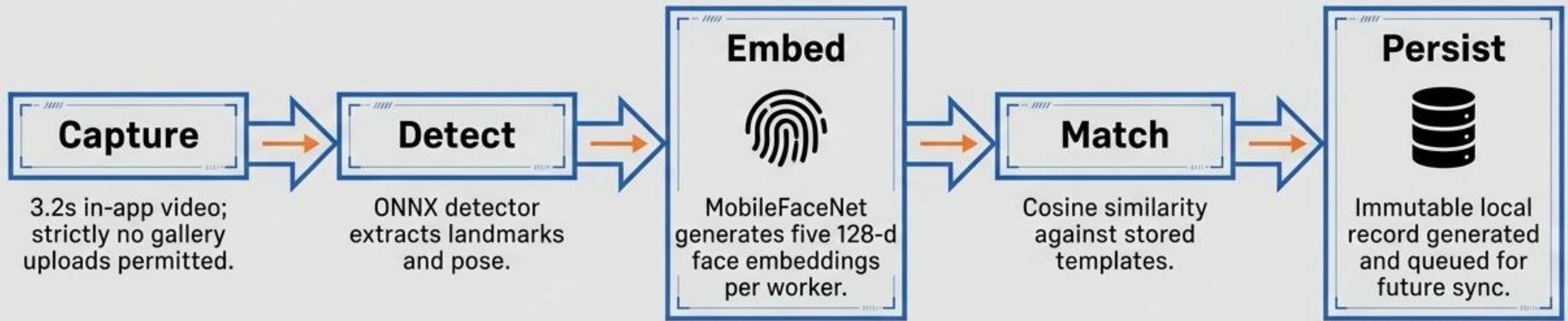
A three-layer cryptographic ecosystem built for zero-connectivity



The Edge Verification Pipeline runs locally in 3.2 seconds



Local Edge Device (Zero Cloud Dependency)



Match Accuracy:
97.28%

Model Footprint:
16.14 MB

Verification Latency:
<1 sec

Minimum Device Spec:
3 GB RAM
(sub-₹8,000 Androids)

Defeating buddy-punching by using time as a cryptographic filter

❌ Static Attack (Photo / Replay)



0% Temporal Change

A high-res printout might pass a basic cosine-similarity match, but across six sampled video frames, it produces zero temporal change. Liveness fails by design.

✅ Dynamic Subject (Live Worker)



Blink
Dip-and-Recovery

Head-Pose
Micro-Fluctuations

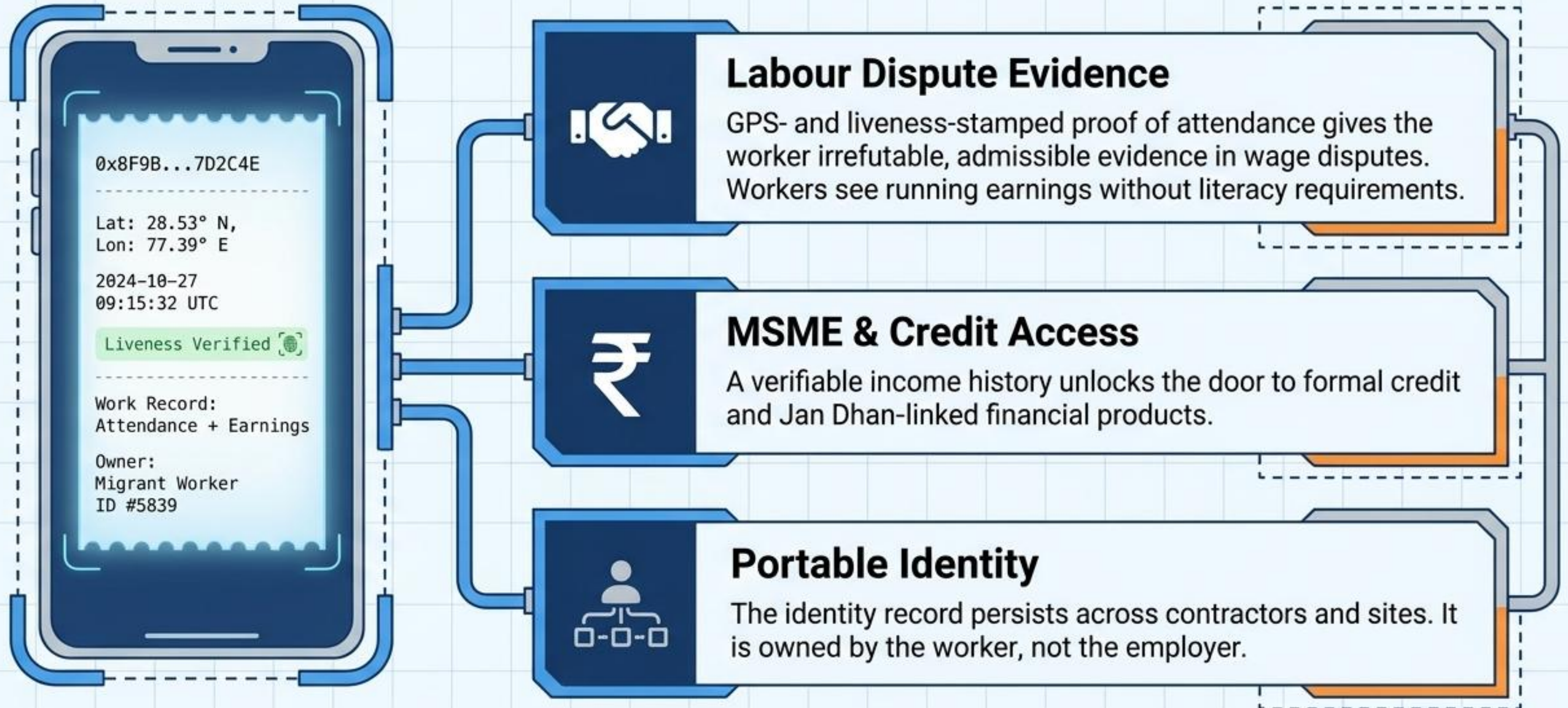
Smile-Probability
Changes

3.2 seconds

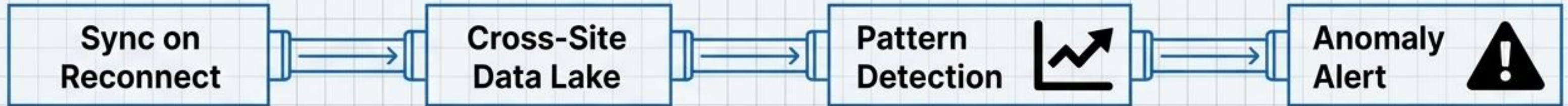
The system fuses three live signals across a 3.2-second window: blink dip-and-recovery curves, continuous head-pose micro-fluctuations, and smile-probability changes.

Result: Attendance logs *only* if identity AND liveness both pass.

The first verifiable, portable work record a migrant labourer has ever owned



The Cloud Anomaly Engine catches ghost workers before payroll runs



The Process: An LLM-powered anomaly engine compares the cryptographically verified digital attendance against the paper musters submitted by supervisors, surfacing discrepancies proactively.

Dashboard Flags

